

Power BI Assignment 1 – Data Transformation, Data Modeling

Transformation Steps :

- ❖ First, I Imported the CSV File Subsequently, I selected “Transform Data” to Load it into Power Query.
- ❖ I then Applied Row limiting and Formatted the data. Following that, I converted the “Customer Name” Field to Proper Case.
- ❖ I Created a New Column by Matching the “City” and “State” Fields, And Generated a Conditional Column Based On The “Profit States” Data.
- ❖ Location New Column

The screenshot shows the Power BI Query Editor with the 'List of Orders' table. The formula bar contains the M code: `Table.AddColumn(#"CustomerName to Proper Case", "Location", each Text.Combine({[City], [State]}, ","), type text)`. The 'Applied Steps' pane on the right shows the sequence: Source, Navigation, Promoted Headers, Order Date Data Type Changed, CustomerName to Proper Case, and Merged City and State. The table view shows columns for Order Date, CustomerName, State, City, and the newly created Location column, which combines City and State with a comma separator.

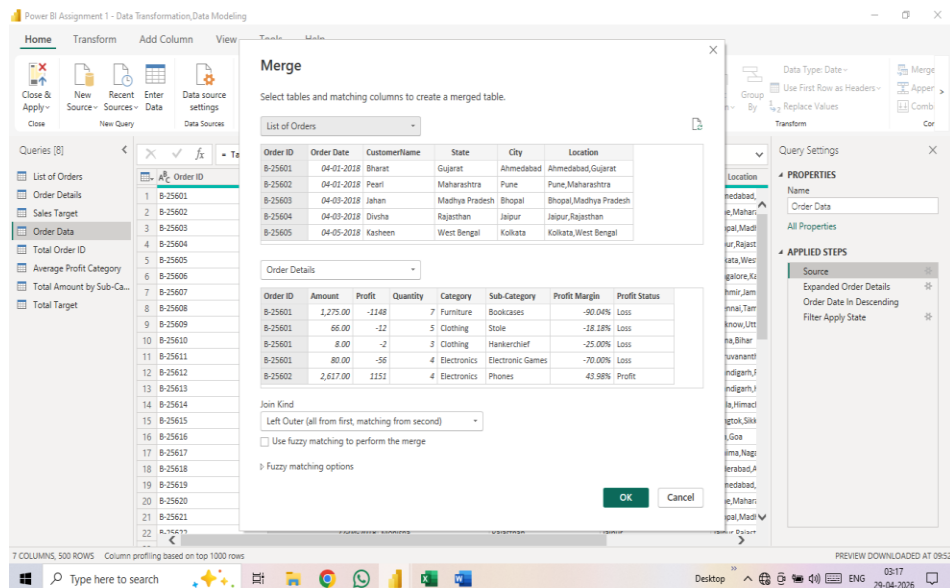
Order Date	CustomerName	State	City	Location
04-01-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
04-01-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
04-01-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
04-01-2018	Dhivha	Rajasthan	Jaipur	Jaipur,Rajasthan
04-01-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
04-06-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
04-06-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
04-08-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
04-09-2018	Itesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
04-09-2018	Yogesh	Bihar	Patna	Patna,Bihar
04-11-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
04-12-2018	Divachand	Punjab	Chandigarh	Chandigarh,Punjab
04-12-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
12-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15-04-2018	Shweta	Sikkim	Gangtok	Gangtok,Sikkim
15-04-2018	Kanak	Goa	Goa	Goa,Goa
17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20-04-2018	Santa	Maharashtra	Pune	Pune,Maharashtra
20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
21-04-2018	Mukesh	Delhi	New Delhi	New Delhi,Delhi

❖ Conditional Column

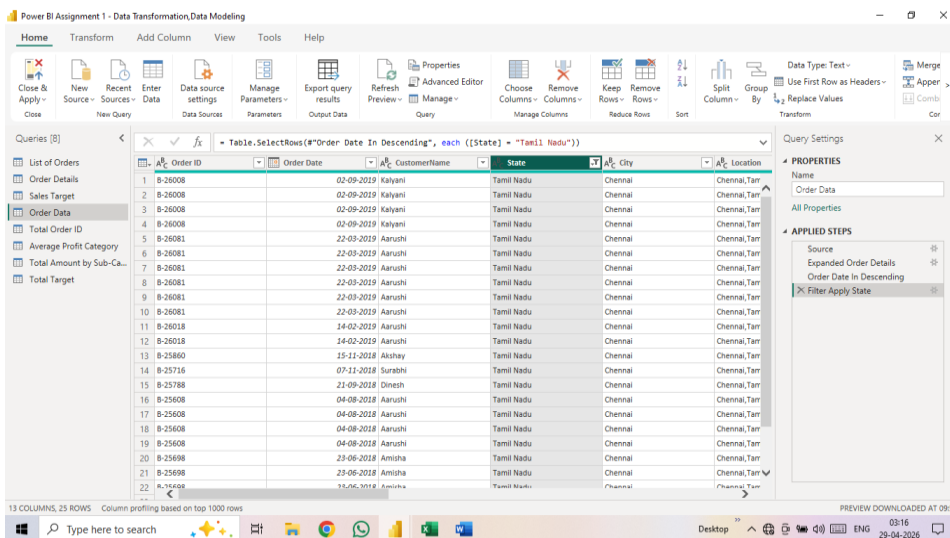
The screenshot shows the Power BI Query Editor with the 'Order Details' table. The formula bar contains the M code: `Table.AddColumn(#"Profit Margin Column Data Type Changed", "Profit Status", each if [Profit] < 100 then "Loss"`. The 'Applied Steps' pane on the right shows the sequence: Source, Navigation, Promoted Headers, Changed Data Type, Amount Column Data Type C..., Profit Margin Column Data Ty..., and Profit Status Conditional Colu... The table view shows columns for Quantity, Category, Sub-Category, Profit Margin, and the newly created Profit Status column, which is set to 'Loss' for profit margins less than 100.

Quantity	Category	Sub-Category	Profit Margin	Profit Status
2249	Furniture	Bookcases	-60.04%	Loss
12	Clothing	Stole	-18.18%	Loss
2	Clothing	Hankchief	-25.00%	Loss
56	Electronics	Electronic Games	-70.00%	Loss
111	Electronics	Phones	-66.07%	Loss
272	Electronics	Phones	-64.15%	Loss
1111	Electronics	Phones	43.98%	Profit
222	Clothing	Saree	37.79%	Break-Even
5	Clothing	Saree	-4.20%	Loss
40	Clothing	Trousers	-4.43%	Loss
30	Furniture	Chairs	-125.00%	Loss
186	Clothing	Kurti	-86.01%	Loss
5	Clothing	Trousers	47.37%	Loss
16	Clothing	Stole	32.79%	Loss
36	Clothing	Stole	33.64%	Loss
1	Clothing	Hankchief	8.33%	Loss
18	Clothing	T-shirt	26.15%	Loss
17	Clothing	T-shirt	3.18%	Loss
5	Clothing	Saree	0.00%	Loss
0	Clothing	Saree	4.60%	Loss
4	Clothing	Stole	61.96%	Loss

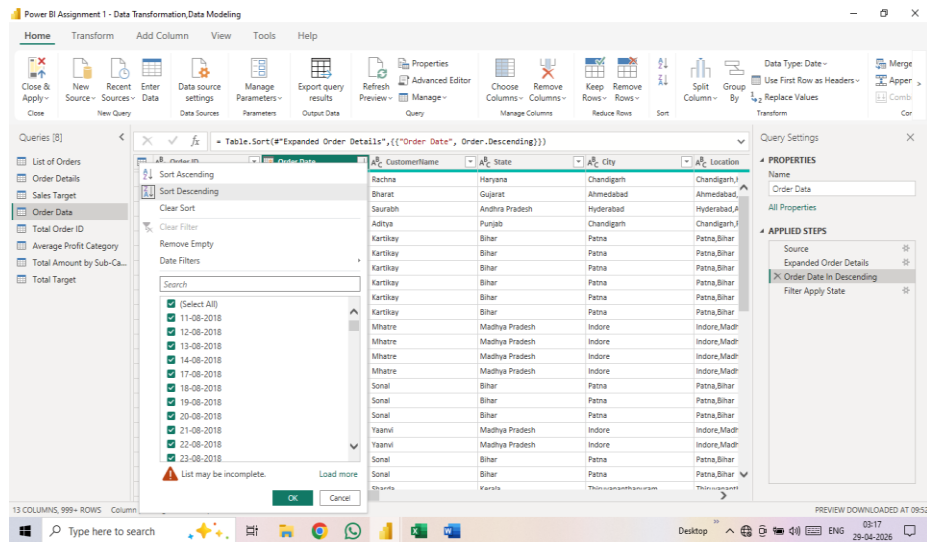
- ❖ Furthermore, I Matched the “List of Orders” and “Order Details” tables to create a Separate, New table.



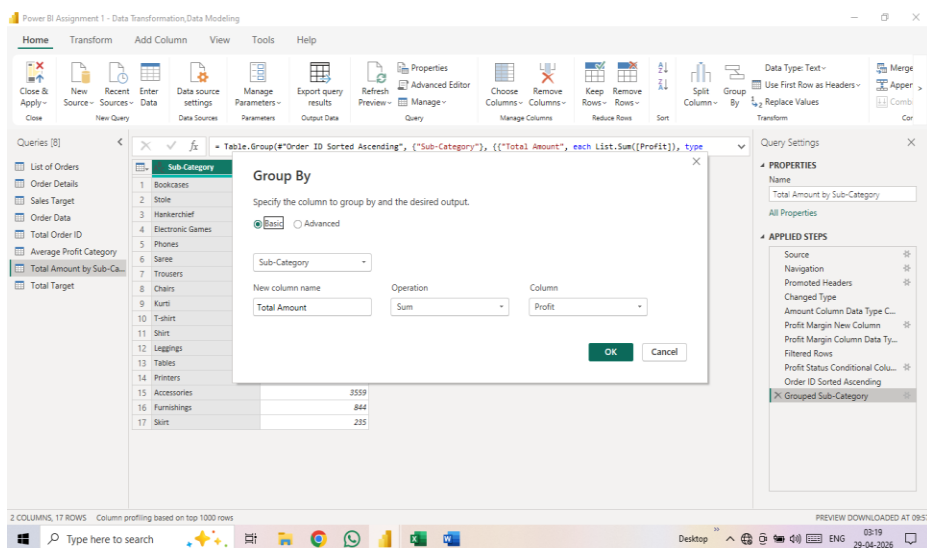
- ❖ To handle Missing values, I Utilized the “Replace Values” and “Remove Duplicates” Options.
- ❖ By Applying Sorting and Filtering, I was able to Segregate and View the data on a State- By -State Basis.



- ❖ Furthermore, I was able to sort the Data to view the order dates in Descending Order.

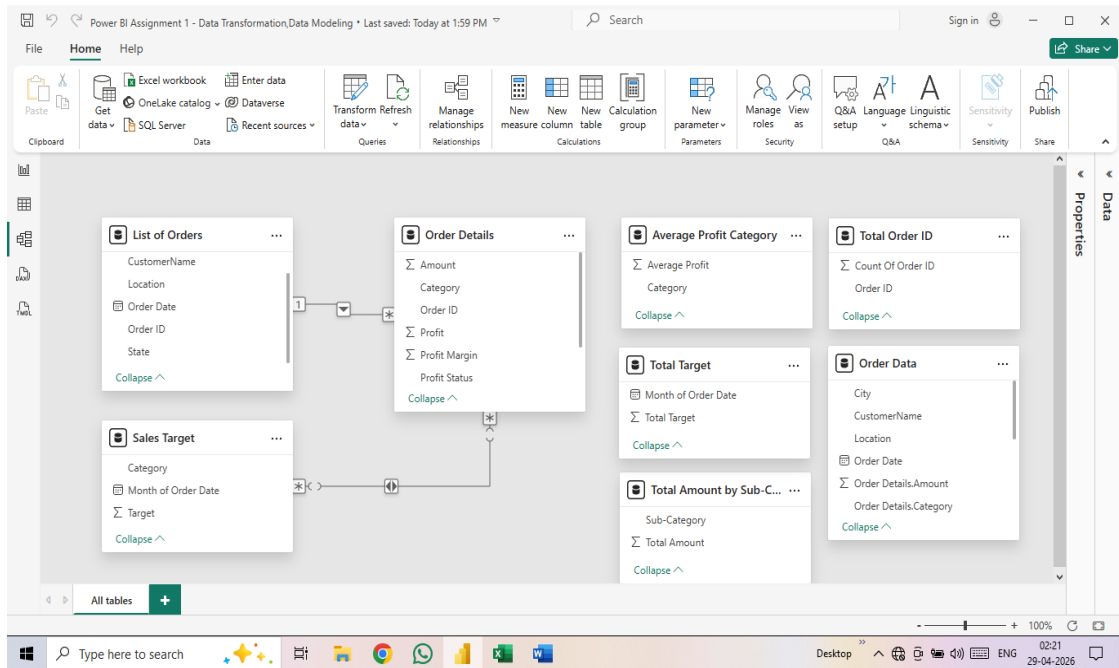


- ❖ Using the “Group By” option, I was able to Determine the total Monthly targets, the Average Total by category, and the total profits By Sub-Category.



Modeling Steps:

- ❖ By tilizing Data modelling, I Was Able to Establish Relationships between Separate tables.
- ❖ Furthermore , by Using the “Order ID” Found in both the “List of Orders” and “Order Details” Table.
- ❖ I was Able to Create a ONE-TO-MANY Relationship.
- ❖ Additionally, I was Able to link the “Sales Target” and “Order Details” tables based on the “Category” Field, Thereby Establishing a MANY-TO-MANY Relationship.



❖ Finally, By Clicking on “Manage Relationship” I Was Able to verify that These Relationships Were Indeed Active.